

FCU3 Fabrication Document

Layer Stack Legend

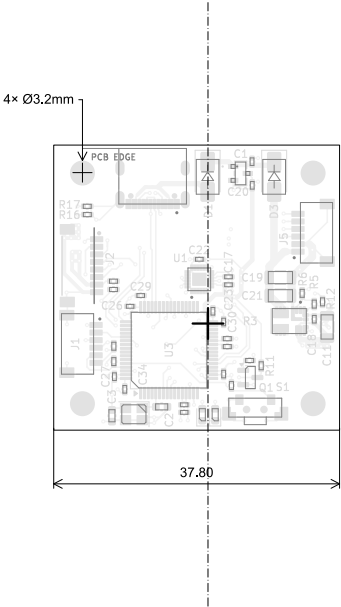
	Material	Layer	Thickness	Dielectric	Type	Gerber
	F,Paste				Paste Mask	
	F,Silkscreen			Direct Printing	Legend	GBR
	F,Mask		0,02mm	Solder Resist	Solder Mask	GBR
	Copper	L1 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
	Prepreg		0,1mm	FR4_7628	Dielectric	
	Copper	L2 (GND)	0,035mm (1,00oz)		Plane	GBR
	Core		1,21mm	FR4	Dielectric	
	Copper	L3 (Sig, PWR)	0,035mm (1,00oz)		Plane	GBR
	Prepreg		0,1mm	FR4_7628	Dielectric	
	Copper	L4 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
	B,Mask		0,02mm	Solder Resist	Solder Mask	GBR
	B,Silkscreen			Direct Printing	Legend	GBR
	B,Paste				Paste Mask	

Total thickness: 1,66mm
Note: external layer thicknesses are specified after plating

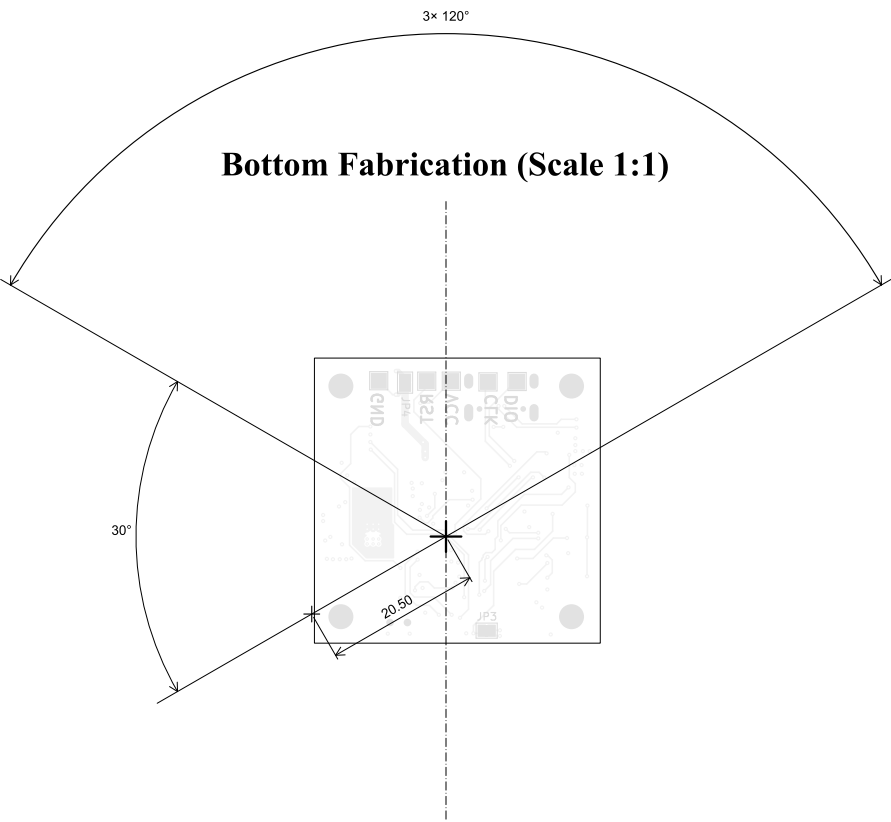
Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0,2032	0,28	L2

Top Fabrication (Scale 1:1)



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All dimensions are in millimeters unless otherwise specified.

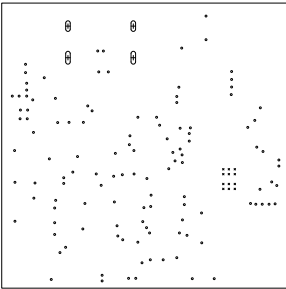
	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 5305e52
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 2 of 10

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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	12	0,20mm (7,87mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
O	120	0,25mm (9,84mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Via
+	4	0,60mm (23,62mils)	PTH	Slot	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
Total 136						

Drill Drawing L1 - L4 (Scale 1:1)



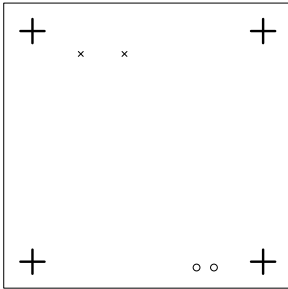
	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 5305c52
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 3 of 10

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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	2	0,85mm (25,59mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
O	2	0,90mm (35,43mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
+	4	3,20mm (125,98mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
Total 8						

Drill Drawing L1 - L4 (Scale 1:1)



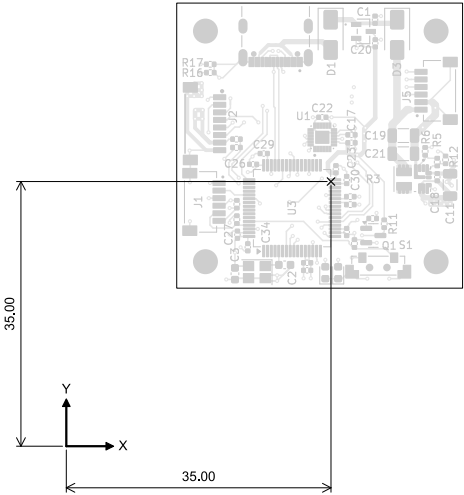
	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 5305c52
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 4 of 10

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Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
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Ref.	Net	X [mm]	Y [mm]
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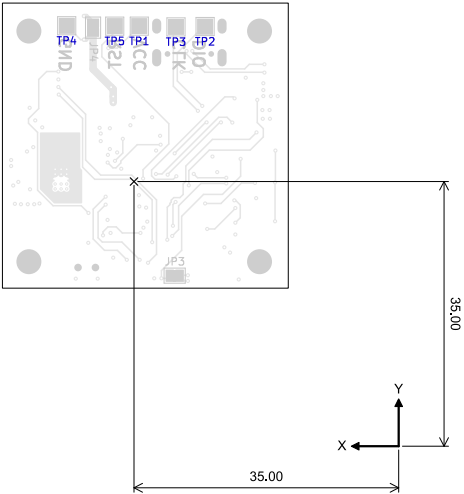
All dimensions are in millimeters unless otherwise specified.

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		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Top Test Points (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 5 of 10

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Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP1	+3V3	19.70	-3.05
TP2	SWDIO	11.00	-3.10
TP3	SWCLK	14.85	-3.15
TP4	DGND	29.30	-2.99
TP5	Net-(U3-NRST)	22.95	-3.05

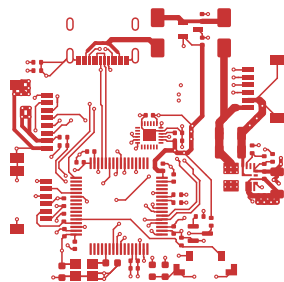


All dimensions are in millimeters unless otherwise specified.

	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 5305c52
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: Bottom Test Points (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
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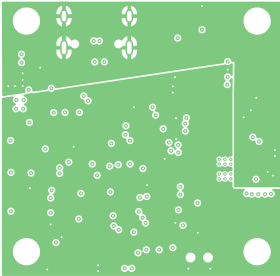
L1 (Sig, PWR) (Scale 1:1)



	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 5305e52
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: L1 (Sig, PWR) (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
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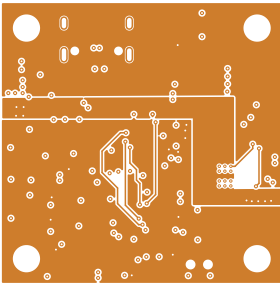
L2 (GND) (Scale 1:1)



	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 5305c52
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: L2 (GND) (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
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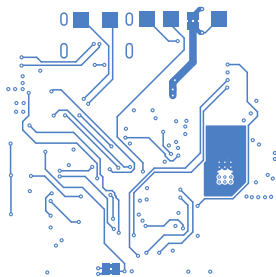
L3 (Sig, PWR) (Scale 1:1)



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		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: L3 (Sig, PWR) (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
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L4 (Sig, PWR) (Scale 1:1)



	Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 5305c52
		Board Name: FCU3		Project Name: Flight Controller 3	
	Sheet Title: L4 (Sig, PWR) (Scale 1:1)	File Name: FCU3.kicad_pcb	Designer: Kaldstrand	Date: 2024-04-13	Revision: + (Unreleased)
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